Homework 09 Solutions

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Choosing an SVM kernel for the South African heart disease data by five-fold cross-validation, and a short hand-written SVM analysis skill applied to the same data.

Use of AI tools. I used Claude (Claude Code) to draft the `R` code and exposition for Question 1 and to run it. Every number below comes from the code output shown in this report, and I checked the numbers quoted in the text against it. For Question 2, I wrote `SKILL.md` by hand without AI assistance; as the assignment requires, an AI agent applied the skill to produce the one-page report, which I then checked.

Question 1: Choosing an SVM kernel

```
suppressWarnings(library(e1071))
heart <- read.csv("data/saheart.csv")
heart$famhist <- ifelse(heart$famhist == "Present", 2, 1)   # Absent = 1, Present = 2
X <- as.matrix(heart[, setdiff(names(heart), "chd")])
y <- factor(heart$chd)
n <- nrow(X); p <- ncol(X)
c(n = n, p = p, chd_rate = mean(heart$chd))
```

```
          n           p    chd_rate
462.0000000   9.0000000   0.3463203
```

```
set.seed(432)                                   # stratified 5 folds, shared by all candidates
fold <- integer(n)
for (cls in levels(y)) {
  idx <- which(y == cls)
  fold[idx] <- sample(rep(1:5, length.out = length(idx)))
}
table(fold, y)
```

```
    y
fold  0  1
   1 61 32
   2 61 32
   3 60 32
   4 60 32
   5 60 32
```

Candidates (19). Linear: $C \in \{0.01, 0.1, 1, 10\}$. Polynomial of degree 2, 3, 4: $C \in \{0.1, 1, 10\}$ each, with the kernel $(\gamma x^\top x' + r)^d$ held at $\gamma = 1/p = 1/9$ and $r = 1$. Radial $\exp(-\gamma\|x - x'\|^2)$: $C \in \{0.1, 1, 10\}$ and $\gamma \in \{0.01, 0.1\}$. All fits use `e1071::svm` with `scale = FALSE`, after standardizing each fold's fitting observations and applying their means and standard deviations to the held-out fold.

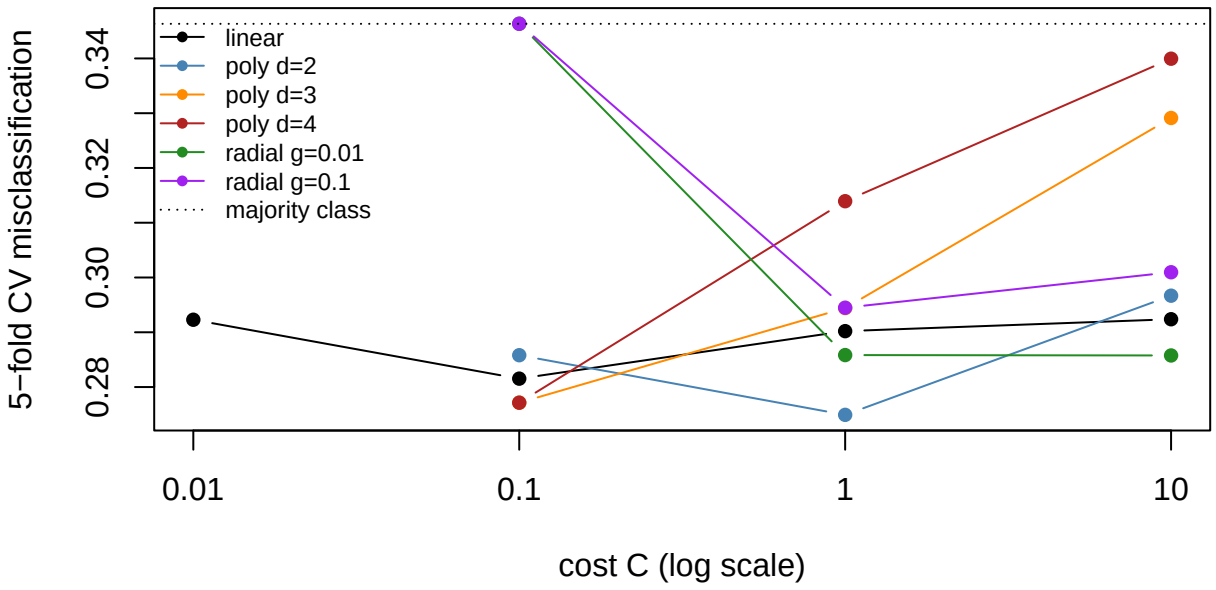
```
cands <- rbind(
  data.frame(kernel = "linear",     degree = NA, gamma = NA,   cost = c(0.01, 0.1, 1, 10)),
  expand.grid(kernel = "polynomial", degree = 2:4, gamma = 1 / p, cost = c(0.1, 1, 10),
              stringsAsFactors = FALSE),
  expand.grid(kernel = "radial",     degree = NA, gamma = c(0.01, 0.1), cost = c(0.1, 1, 10),
              stringsAsFactors = FALSE))
stopifnot(nrow(cands) <= 20)

fit_one <- function(Xtr, ytr, Xte, cand) {
  args <- list(x = Xtr, y = ytr, kernel = cand$kernel, cost = cand$cost, scale = FALSE)
  if (cand$kernel == "polynomial") args <- c(args, degree = cand$degree, gamma = cand$gamma, coef0 = 1)
  if (cand$kernel == "radial")     args <- c(args, gamma = cand$gamma)
  predict(do.call(svm, args), Xte)
}

err <- matrix(NA, nrow(cands), 5)
for (k in 1:5) {
  tr <- fold != k
  mu <- colMeans(X[tr, ]); s <- apply(X[tr, ], 2, sd)
  Xtr <- scale(X[tr, ], mu, s); Xte <- scale(X[!tr, ], mu, s)
  for (j in seq_len(nrow(cands)))
    err[j, k] <- mean(fit_one(Xtr, y[tr], Xte, cands[j, ]) != y[!tr])
}
cands$cv_error <- rowMeans(err)
cands$se <- apply(err, 1, sd) / sqrt(5)
cands$label <- with(cands, ifelse(kernel == "linear", "linear",
                   ifelse(kernel == "polynomial", paste0("poly d=", degree),
                          paste0("radial g=", gamma))))
knitr::kable(cands[, c("kernel", "degree", "gamma", "cost", "cv_error", "se")],
             digits = 3, row.names = FALSE)
```

kernel	degree	gamma	cost	cv_error	se
linear	NA	NA	0.01	0.292	0.011
linear	NA	NA	0.10	0.282	0.016
linear	NA	NA	1.00	0.290	0.017
linear	NA	NA	10.00	0.292	0.018
polynomial	2	0.111	0.10	0.286	0.012
polynomial	3	0.111	0.10	0.277	0.009
polynomial	4	0.111	0.10	0.277	0.011
polynomial	2	0.111	1.00	0.275	0.009
polynomial	3	0.111	1.00	0.294	0.008
polynomial	4	0.111	1.00	0.314	0.017
polynomial	2	0.111	10.00	0.297	0.018
polynomial	3	0.111	10.00	0.329	0.013
polynomial	4	0.111	10.00	0.340	0.017
radial	NA	0.010	0.10	0.346	0.001
radial	NA	0.100	0.10	0.346	0.001
radial	NA	0.010	1.00	0.286	0.015
radial	NA	0.100	1.00	0.294	0.011
radial	NA	0.010	10.00	0.286	0.008
radial	NA	0.100	10.00	0.301	0.017

```r
groups <- unique(cands$label)
cols <- c("black", "steelblue", "darkorange", "firebrick", "forestgreen", "purple")
par(mar = c(4, 4, 1, 1))
plot(NA, xlim = range(log10(cands$cost)), ylim = range(cands$cv_error), xaxt = "n",
     xlab = "cost C (log scale)", ylab = "5-fold CV misclassification")
axis(1, at = log10(unique(cands$cost)), labels = unique(cands$cost))
for (g in seq_along(groups)) {
  s <- cands[cands$label == groups[g], ]
  lines(log10(s$cost), s$cv_error, type = "b", pch = 16, col = cols[g])
}
abline(h = mean(heart$chd), lty = 3)              # always predict "no CHD"
legend("topleft", legend = c(groups, "majority class"), col = c(cols, "black"),
       lty = c(rep(1, length(groups)), 3), pch = c(rep(16, length(groups)), NA), cex = 0.75, bty = "n")
```




```r
best <- cands[which.min(cands$cv_error), ]
one_se <- cands[cands$cv_error <= best$cv_error + best$se,   # within one SE of the best
                c("kernel", "degree", "gamma", "cost", "cv_error")]
best[, c("kernel", "degree", "gamma", "cost", "cv_error", "se")]
```

```
      kernel degree     gamma cost  cv_error         se
8 polynomial      2 0.1111111    1 0.2749182 0.00898764
```

```r
one_se[order(one_se$cv_error), ]
```

```
      kernel degree     gamma cost  cv_error
8 polynomial      2 0.1111111  1.0 0.2749182
6 polynomial      3 0.1111111  0.1 0.2771388
7 polynomial      4 0.1111111  0.1 0.2771388
2     linear     NA        NA  0.1 0.2815334
```

Preferred model. The lowest CV error is the quadratic polynomial kernel with $C = 1$ (0.275, SE 0.009). It is not clearly better than its neighbors, though. Cubic and quartic kernels with $C = 0.1$ (0.277) and the linear kernel with $C = 0.1$ (0.282) all fall within one standard error of it, and every candidate is only a few points better than always predicting "no CHD" (0.346). Raising C hurts the higher-degree polynomials (0.340 for degree 4 at $C = 10$), which

points to overfitting. The radial kernel with $C = 0.1$ collapses to the majority class, and the best radial settings (0.286) do not beat the linear kernel. By the one-standard-error rule I prefer the **linear kernel with** $C = 0.1$. It is the simplest model whose error is statistically indistinguishable from the best. Its boundary is also a weighted sum of the standardized predictors, which is easy to interpret. The extra flexibility of nonlinear kernels does not pay off with 462 observations and nine predictors. If the goal were only the smallest CV error, the quadratic kernel with $C = 1$ would be the choice.

Question 2: SVM analysis skill

Pending: one-page report produced by invoking my hand-written skill, followed by the full `SKILL.md`.